

Problem Set 8 Question 1

(a) IS-MPR $y_t = y_t^* - \alpha(\beta\pi - 1)(\pi_t - \pi^*) + \varepsilon_t^y$

PC $\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$

Accordingly we write

$$\Delta y_t = \Delta y_t^* - \alpha(\beta\pi - 1)\Delta\pi_t + \alpha(\beta\pi - 1)\Delta\pi^* + \Delta\varepsilon_t^y$$

$$\Delta\pi_t = \Delta\pi_t^e + \gamma\Delta y_t - \gamma\Delta y_t^* + \Delta\varepsilon_t^\pi$$

When we want to know the effects of $\Delta\varepsilon_t^y$ on Δy_t & $\Delta\pi_t$, we set

$$\Delta y_t^* = \Delta\pi^* = \Delta\pi_t^e = 0$$

$$\Delta y_t + \alpha(\beta\pi - 1)\Delta\pi_t = \Delta\varepsilon_t^y$$

$$-\gamma\Delta y_t + \Delta\pi_t = 0$$

You can solve using Cramer's rule or by subⁿ

$$\begin{bmatrix} 1 & \alpha(\beta\pi - 1) \\ -\gamma & 1 \end{bmatrix} \begin{bmatrix} \Delta y_t \\ \Delta\pi_t \end{bmatrix} = \begin{bmatrix} \Delta\varepsilon_t^y \\ 0 \end{bmatrix}$$

Then,

$$\Delta y_t = \frac{\begin{vmatrix} \Delta\varepsilon_t^y & \alpha(\beta\pi - 1) \\ 0 & 1 \end{vmatrix}}{1 + \alpha\gamma(\beta\pi - 1)} = \frac{1}{1 + \alpha\gamma(\beta\pi - 1)} \cdot \Delta\varepsilon_t^y = \theta \Delta\varepsilon_t^y$$

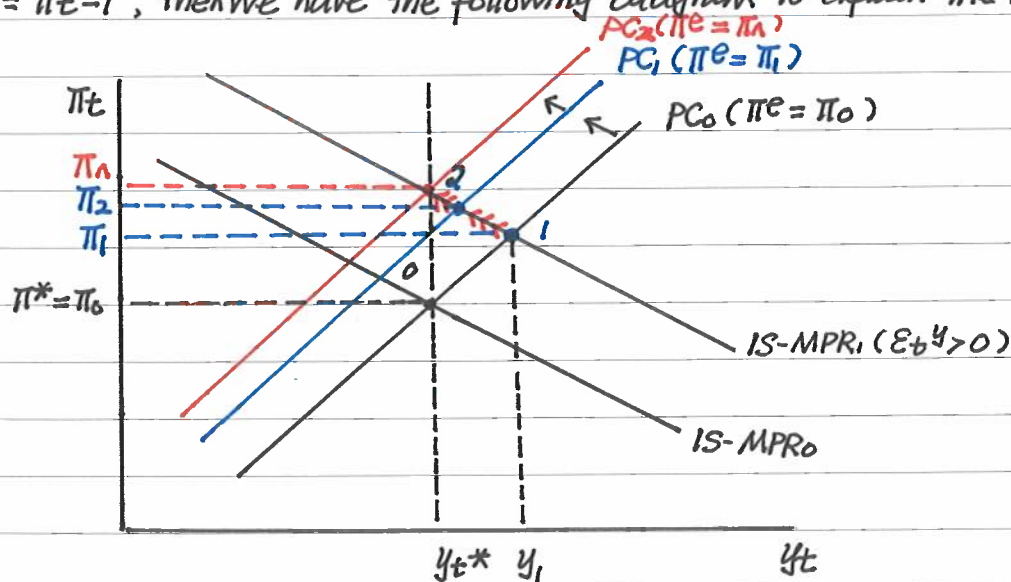
$$\Delta\pi_t = \frac{\begin{vmatrix} 1 & \Delta\varepsilon_t^y \\ -\gamma & 0 \end{vmatrix}}{1 + \alpha\gamma(\beta\pi - 1)} = \frac{\gamma}{1 + \alpha\gamma(\beta\pi - 1)} \cdot \Delta\varepsilon_t^y = \gamma\theta \Delta\varepsilon_t^y$$

When $\Delta\varepsilon_t^y > 0$, then $\Delta y_t = \theta \cdot \Delta\varepsilon_t^y > 0$ and $\Delta\pi_t = \gamma\theta \Delta\varepsilon_t^y > 0 \therefore y_t \uparrow, \pi_t \uparrow$

This solution should be consistent with the diagram and intuition (Refer PS 7)

You should consider solving Q2 and Q3 using similar technique.

- (b) Assuming that expectation of inflation is formed based on last period inflation $\pi_t^e = \pi_{t-1}$, then we have the following diagram to explain the adjustment



Note that every period π^e adjusts according to last period π and PC shifts gradually from PC_0 to PC_n with $y_t \downarrow$ back to y_t^* and $\pi_t = \pi^e = \pi_n$.

Question 4

(a) MPR is given by $i_t = r^* + \pi^* + \beta\pi(\pi_t - \pi^*)$

Accordingly $\Delta i_t = \Delta r^* + \Delta \pi^* + \beta\pi \Delta \pi_t - \beta\pi \Delta \pi^*$

Holding all else constant, $\Delta r^* = \Delta \pi_t = 0$. Then,

$$\Delta i_t = -\beta\pi \cdot \Delta \pi_t^*$$

This means that to reduce i_t to both the economy

$$\Delta i_t = -\beta\pi \cdot \Delta \pi_t^* < 0 \quad \therefore \Delta \pi_t^* > 0 \quad \therefore \pi^* \uparrow$$

i.e. to expand the economy with lower i_t , central bank must accept/set higher inflation target, π^* .

Therefore

IS-MPR $\Delta y_t = (\Delta y_t^*) - \alpha(\beta\pi - 1)\Delta \pi_t + \alpha(\beta\pi - 1) \cdot \Delta \pi^* + (\Delta \varepsilon_t^y)$

PC $\Delta \pi_t = (\Delta \pi_t^e) + \gamma \Delta y_t - (\gamma \Delta y_t^*) + (\Delta \varepsilon_t^\pi)$

Let $\Delta y_t^* = \Delta \varepsilon_t^y = \Delta \pi_t^e = \Delta \varepsilon_t^\pi = 0$, we have

$$\Delta y_t = -\alpha(\beta\pi - 1)\Delta \pi_t + \alpha(\beta\pi - 1)\Delta \pi^*$$

$$\Delta \pi_t = \gamma \Delta y_t$$

You can solve using Cramer's rule or by subⁿ

$$\begin{bmatrix} 1 & \alpha(\beta\pi - 1) \\ -\gamma & 1 \end{bmatrix} \begin{bmatrix} \Delta y_t \\ \Delta \pi_t \end{bmatrix} = \begin{bmatrix} \alpha(\beta\pi - 1) \cdot \Delta \pi^* \\ 0 \end{bmatrix}$$

Then,

$$\Delta y_t = \frac{\alpha(\beta\pi - 1)}{1 + \alpha\gamma(\beta\pi - 1)} \cdot \Delta \pi^* > 0$$

$$\Delta \pi_t = \frac{\alpha\gamma(\beta\pi - 1)}{1 + \alpha\gamma(\beta\pi - 1)} \cdot \Delta \pi^* > 0$$

When $\Delta \pi^* > 0$, then $\Delta y_t > 0$ and $\Delta \pi_t > 0 \quad \therefore y_t \uparrow$ and $\pi_t \uparrow$.

This solution should be consistent with the diagram and intuition (Refer PS7).

(b) Refer to solutions for PS8 Q1(b)

(c) Based on (b), reduction in i_t (expansionary monetary policy) cannot have a permanent effect on y_t in the IS-MPR framework because π_t^e will adjust and eventually $\pi_t^e = \text{new inflation target}$ and $y_t = y_t^*$.